

Bachelor Thesis

Proactive AI moderation for goal-oriented remote meetings: Design and acceptance of a context-sensitive system to support convergent decision-making processes

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Contents

1	Introduction	5
1.1	Background and Motivation	6
1.1.1	Inefficiencies in remote meetings: Challenges in convergent decision-making processes	6
1.1.2	Potential of AI-driven facilitation to enhance goal orientation . . .	6
1.2	Research Problem and Gap	7
1.2.1	Limitations of existing moderation tools in remote settings	7
1.2.2	Need for context-sensitive, proactive AI systems	7
1.3	Research Objectives and Questions	8
1.3.1	RQ1: Facilitating strategies and challenges in goal-oriented remote meetings	8
1.3.2	RQ2: Design requirements for an AI-based moderation system . . .	8
1.3.3	RQ3: User acceptance among knowledge workers	9
2	Theoretical Foundations	10
2.1	Remote Meetings	10
2.1.1	Meetings in Modern Work: Trends and Context	10
2.1.2	Key challenges: Off-topic discussions, unequal participation, lack of structure	11
2.2	Facilitation Strategies for Remote Collaboration (RQ1)	12
2.2.1	Empirical evidence on effective techniques (e.g., timeboxing, role assignment)	12
2.2.2	Technological and social dynamics in virtual teams	13
2.3	Decision-Making	13
2.3.1	Definitions and distinctions (convergent vs. divergent processes) . .	13
2.3.2	Divergent and Convergent Thinking in Meetings	13
2.4	AI in Meeting Support Systems (RQ2)	13
2.4.1	Overview of existing Meeting AI Tools	13

2.4.2	AI-Agents in General	14
2.4.3	Building blocks for AI-Agents	14
2.4.4	Frameworks to design LLM-based Meeting AI-Agents	15
2.4.5	Proactive AI-Interactions: Timing and triggers for Human-AI-Interaction- Systems	15
2.4.6	Theoretical frameworks: Human-AI Interaction (HAI), Explainable AI (XAI)	16
2.5	Technology Acceptance and Usability	16
2.5.1	Technology Acceptance Model (TAM)	17
2.5.2	System Usability Scale (SUS) and psychological effects	17
3	Empirical Analysis of Facilitating Strategies (RQ1)	18
3.1	Methodology: Expert Interviews	18
3.1.1	Sampling: Purposeful selection of 5 facilitating experts	19
3.1.2	Data Collection: Conduction of semi-structured interviews (30 min)	19
3.1.3	Analysis: Thematic coding	19
3.2	Results: Facilitating Strategies and Challenges	20
3.2.1	Comparison with literature (Chapter 2)	20
3.3	Implications for AI System Design	20
3.3.1	Derived requirements for RQ2	20
4	System Design and Implementation (RQ2)	21
4.1	Requirements Analysis	21
4.1.1	Functional Requirements	21
4.1.2	Non-Functional Requirements	22
4.2	System Architecture	22
4.2.1	Components: Transcription, Analysis, and Intervention Modules	22
4.2.2	Technology Stack: NLP Libraries, Backend/Frontend Frameworks	22
4.3	Prototype Development	22
4.3.1	Implementation Details	22
4.3.2	Limitations of the Prototype	22
5	Empirical Validation (RQ2 and RQ3)	23
5.1	Methodology: Usability Study	23
5.1.1	Study Design	23
5.1.2	Sample	23
5.1.3	Data Collection	23



5.2	Results	23
5.2.1	Usability (SUS Scores and Analysis)	23
5.2.2	Acceptance (TAM: Perceived Usefulness, Ease of Use, Psychological Effects)	23
5.3	Discussion	23
5.3.1	Implications for System Design	23
5.3.2	Theoretical Contributions	23
5.3.3	Practical Impact for Organizations	23
6	Conclusion	24
6.1	Summary of Findings	24
6.2	Limitations	24
6.3	Future Work	24
7	Appendix	28
7.1	Semi-Structured Interview Guide for Expert Interviews with Facilitators . .	28



1 Introduction

Remote meetings have become a cornerstone of modern collaboration. Since the start of the COVID-19 pandemic, the use of video conferencing (VC) industry has risen dramatically [1]. Hardly any meeting takes place without at least one remote participant [Source] Yet they often suffer from inefficiencies, particularly in convergent decision-making processes such as prioritization, problem-solving, and consensus-building. Research by Atlassian indicates that up to 72% of meetings are considered ineffective, with common issues including off-topic discussions, unequal participation, and a lack of structure. [2]

Although traditional facilitation strategies are effective, their practical implementation is limited by the fact that not every team has permanent access to a trained facilitator. Professional training and external consulting, while valuable, are costly and not continuously available.

The potential of AI-driven facilitation to address these gaps lies in its ability to provide real-time, context-sensitive support. Unlike static tools (e.g., timers or agendas), proactive AI systems can analyze conversations, detect deviations from predefined goals, and intervene with constructive suggestions—thereby enhancing goal orientation and reducing cognitive load on human moderators. Early adopters of AI in meetings, such as Otter.ai (real-time transcription) and Microsoft Teams’ AI recaps, demonstrate the feasibility of automated support, but these tools lack proactive, adaptive moderation tailored to convergent processes [Microsoft, 2024; Otter.ai, 2023]

1.1 Background and Motivation

1.1.1 Inefficiencies in remote meetings: Challenges in convergent decision-making processes

Convergent decision-making in remote meetings faces unique obstacles:

- Off-topic discussions: Studies show that 30–50% of meeting time is spent on irrelevant tangents, derailing progress [Rogelberg et al., 2006]. [3]
- Unequal participation: Dominant voices can overshadow quieter team members, leading to groupthink and suboptimal outcomes [Nunamaker et al., 1991].
- Lack of structure: Without clear facilitation, meetings often lack timeboxing, role assignment, or explicit decision criteria, reducing efficiency [Schwartz, 2016].

Traditional solutions—such as hiring professional facilitators or training team members—are cost-prohibitive or scalability-limited. For example, a certified facilitator may charge €100–300/hour [International Association of Facilitators, 2024], making continuous support impractical for most organizations. Furthermore, not every team has permanent access to trained personnel, and ad-hoc facilitation often fails to address systemic issues like dysfunctional team dynamics or blind spots in group perception [Edmondson, 1999].

1.1.2 Potential of AI-driven facilitation to enhance goal orientation

AI can bridge this gap by:

- Automating real-time analysis: Natural Language Processing (NLP) can transcribe and semantically analyze discussions to identify deviations from the meeting’s goal [Bommasani et al., 2021].
- Providing adaptive interventions: Context-sensitive prompts (e.g., “Let’s return to the agenda: prioritizing the top 3 solutions”) can refocus conversations without human intervention.
- Democratizing participation: AI can nudge underrepresented voices (e.g., “Alex, you haven’t shared your perspective yet—would you like to add anything?”) or summarize key points to ensure alignment.

Pilot studies on AI moderation tools (e.g., Humley’s AI meeting assistant) report 20–30% reductions in meeting duration and improved participant satisfaction [Humley, 2023]. However, these tools often lack proactivity—reacting to issues rather than preventing them—and customization for specific use cases like convergent decision-making.

1.2 Research Problem and Gap

1.2.1 Limitations of existing moderation tools in remote settings

Current tools fall short in three key areas:

- **Passive functionality:** Most AI tools (e.g., Zoom AI Companion, Google Meet’s recaps) focus on post-meeting summaries or transcription, not real-time moderation [Zoom, 2024; Google, 2023].
- **Generic interventions:** Solutions like Miro’s AI facilitator provide template-based guidance but fail to adapt to the meeting’s context or user preferences [Miro, 2024].
- **Lack of empirical validation:** Few studies evaluate user acceptance of proactive AI moderation, particularly in high-stakes decision-making scenarios [Benbasat and Barki, 2007].

1.2.2 Need for context-sensitive, proactive AI systems

To address these gaps, this thesis proposes a proactive AI moderation system that:

- Detects thematic deviations in real time using NLP and meeting goal tracking.
- Intervenes constructively with personalized, non-intrusive prompts (e.g., private messages to individuals or group reminders).
- Validates design choices through user studies focusing on usability (SUS) and acceptance (TAM).

Such a system could reduce meeting fatigue, improve decision quality, and lower the barrier to effective facilitation—but its success hinges on user trust and perceived usefulness, which remain underexplored in literature.

1.3 Research Objectives and Questions

This thesis addresses the inefficiencies in goal-oriented remote meetings through a structured approach that integrates three complementary research objectives. The first objective (RQ1) investigates facilitating strategies and challenges in remote meetings to establish a foundation based on real-world practices and pain points. The second objective (RQ2) focuses on designing an AI-based moderation system that translates these insights into technical requirements, ensuring the system is context-aware and goal-oriented. The third objective (RQ3) evaluates user acceptance among knowledge workers.

The work bridges the gap between theoretical facilitation techniques and practical AI-driven solutions, ensuring that the system is both technically robust and user-centric.

1.3.1 RQ1: Facilitating strategies and challenges in goal-oriented remote meetings

Which facilitating strategies are suitable for the goal-oriented management of remote meetings with convergent objectives (such as decision-making, prioritization, problem-solving)? What specific challenges typically arise in this context?

The first research question aims to identify effective strategies for managing convergent remote meetings, such as decision-making, prioritization, and problem-solving, while also analyzing the specific challenges that arise. To achieve this, the thesis conducts expert interviews with certified facilitators, agile coaches, and team leads to gather empirical insights. It also analyzes literature on facilitation techniques, including timeboxing and role assignment, as well as the dynamics of remote collaboration. The findings are compared with existing research to derive best practices and identify pain points, such as off-topic discussions and unequal participation, which AI could potentially address.

1.3.2 RQ2: Design requirements for an AI-based moderation system

How should an AI-based moderation system be designed that autonomously detects thematic deviations based on a predefined meeting goal and real-time transcript analysis and promotes goal orientation through constructive, context-sensitive interventions?

The second research question focuses on developing a proactive AI system that detects thematic deviations in real time and intervenes contextually to maintain goal orientation. This objective is addressed by defining functional requirements, such as real-time transcript analysis and adaptive interventions, and non-functional requirements, including privacy, latency, and scalability. A system architecture is then designed with components for transcription, NLP-based analysis, and intervention logic, incorporating mechanisms for private nudges or group reminders. A prototype is implemented using NLP libraries like Hugging Face Transformers, and its technical feasibility is evaluated.

1.3.3 RQ3: User acceptance among knowledge workers

How do knowledge workers in remote teams evaluate the usefulness, usability, and psychological effects of such an AI moderation tool, and what factors influence its acceptance?

The third research question assesses how knowledge workers perceive the usefulness, usability, and psychological impact of the AI moderation system, as well as the factors influencing its adoption. A usability study is conducted with remote teams, using frameworks such as the System Usability Scale and the Technology Acceptance Model. Quantitative data, including SUS scores and efficiency metrics, are collected alongside qualitative feedback from interviews that explore perceptions of control, trust, and pressure. The analysis also examines psychological effects, such as relief versus intrusion, and adoption barriers, including resistance to automation and privacy concerns.

2 Theoretical Foundations

This chapter presents the theoretical foundations and the current state of research. These form the basis for the subsequent work. Based on a systematic literature review, five key challenges in the context of meetings were identified (2.1.2). These serve as the starting point for the in-depth analysis in the following expert interviews (see Chapter 3) and provide indicators of what to focus on during the subsequent requirements gathering process (see Chapter 4.1).

2.1 Remote Meetings

Definition Meeting (Abgrenzung zu WS und Meetings -> Eigentlich das gleiche)

Was sind remote Meetings?

2.1.1 Meetings in Modern Work: Trends and Context

The Doolde Report "State of Meetings 2023" finds that 40% of meetings are scheduled 1–4 weeks in advance (55% in North America). In contrast, the Mirsooft report reveals that most meetings (57%) are ad hoc, without formal invitations. [4]

Welche Arten von Meetings gibt es?

2.1.2 Key challenges: Off-topic discussions, unequal participation, lack of structure

A clear issue with goal-oriented meetings is the absence or unclear communication of objectives. It seems plausible that a significant number of meetings are scheduled without a defined goal or agenda (as discussed in section 2.1.1). A lack of structure in meetings is therefore one of the key challenges. While an agenda can serve as the foundation for such structure, it alone does not guarantee a successful meeting. Meeting goals are often dynamic and may need to be adjusted as the meeting progresses, due to the associated uncertainties [3, chap 4.1]. Although the agenda can outline a path to achieving the overall meeting goal [5, chap 2.1], without linked sub-goals, it guarantees neither the success of the meeting nor a timely conclusion. [3, chap 4.1] Discussions can nevertheless get out of hand if participants do not know what to prioritize. [3, chap 4.1] An agenda was identified by many respondents (67%) as a key element for successful meetings in Doodle's State of Meetings Report 2019. Only the setting of clear meeting objectives (72%) was cited even more frequently. [6, p. 11]

Based on this, the following summary challenge statement was derived:

Challenge 1: Clear goal-setting and structure. Goal-oriented meetings frequently lack clearly communicated objectives and a reliable structure, so that discussions lose focus and the meeting neither succeeds nor concludes on time.

Chen et al. noted, as a secondary finding, that individuals in higher hierarchical positions tended to intervene more frequently and maintain focus on meeting objectives when a clear list of goals was provided. Conversely, those in lower hierarchical positions felt less obligated to intervene, even when they observed deviations from the main topic. [3] Doodle's State of Meetings Report 2019 identified a correlation between salary and self-perceived participation in meetings. Respondents with higher salaries were more likely to describe themselves as active participants. [6, p. 6] This may stem from inexperience, hierarchical constraints, group dynamics, or a lack of error culture, where dissent risks loss of face. Low participation can narrow solution spaces, yield monotonous outcomes, and reduce meeting effectiveness.

Dysfunctional communication—such as complaining, criticizing, and digressing—reduces effectiveness. [7, p. 130]

-> Unausgeglichen / geringe participation -> Unvorhergesehen veränderungen im Meeting (falsch Annahmen, meetingziel ändert sich) -> technische Probleme: Instabile internetverbindungen, Störungen etc. (nicht im scope dieser arbeit aber dennoch relevant)

Graph der zeigt das sich diese challenges alle gegenseitig beinflussen können. (z.B. wenn nicht nicht alle ansichten geführt wurde, kann es später dazu kommen das wenn man eigentlich eine entscheidung treffen möchte (convergent thinking) wieder zurück in divergent thinking fällt, was zu zirkulären diskussionen führen kann.)

Umgekehrt, kann ein zu langes verweilen in der konvergenz zone dazu führen, dass keine entscheidung getroffen wird und kein fortschritt wahrgenommen wird.

Kultur und hierarchie beeinflussen ebenfalls die offen teilnehmer ideen teilen und wer welche ideen wie bewertet.

2.2 Facilitation Strategies for Remote Collaboration (RQ1)

Recherche was sind eigentlich die aufgaben eines Facilitators? was macht ein facilitator?

2.2.1 Empirical evidence on effective techniques (e.g., timeboxing, role assignment)

The book *Facilitation Fundamentals* by AJ&Smart [8], one of the leading companies in the field of facilitation and facilitator training, introduces the 4C framework. This framework classifies methods that can be used in meetings or workshops into four key categories:

- Collect: Suitable for systematically gathering and visualizing information within a team.
- Choose: Narrows down a large number of ideas or content to a few selected elements.
- Create: Uses design thinking methods to develop new approaches to solutions.
- Commit: Ensures the team's binding agreement on the decisions made and the results achieved.

As outlined in the chapter “Convergent and Divergent Thinking,” a clear distinction between the divergent and convergent phases is crucial. The 4C framework, which many facilitators rely on, enables the implementation of these best practices.

2.2.2 Technological and social dynamics in virtual teams

2.3 Decision-Making

2.3.1 Definitions and distinctions (convergent vs. divergent processes)

2.3.2 Divergent and Convergent Thinking in Meetings

Divergent thinking is the process of generating new ideas and concepts [TODO: QUELLE]. Convergent thinking is the process of evaluating and selecting the best ideas from a set of options. [TODO: QUELLE]

According to Anne Manning, generating new ideas and evaluating them are two fundamentally different processes. However, she says that in meetings, people often try to carry out both steps at the same time. [9, min 02:30]

Immediately evaluating and weighing an idea hinders the creative process of generating ideas.

2.4 AI in Meeting Support Systems (RQ2)

2.4.1 Overview of existing Meeting AI Tools

There is a wide variety of AI-based tools that provide support for meetings. Most of these tools focus on transcribing meeting content. Well-known examples include fireflies.ai [10], Otter.ai [11], and Jamie [12]. AI transcription features are also integrated into established meeting platforms such as Google Meet [13], Zoom [14], and Microsoft Teams. [15] More advanced features include the automatic extraction of action items. Depending on the tool, these features are available in real time or only after the meeting has ended. Microsoft Teams specifically offers a feature called “Facilitator,” [16] which provides additional capabilities beyond the features already mentioned (transcription, identification of action

items): For example, it can edit the agenda if instructed to do so by a user with the appropriate permissions. It also displays a timeline to track topics and the flow of the meeting, and sends reminders to manage time. (Once halfway through the meeting and again shortly before it ends.) These interactions take place via the integrated chat. Google Meet, for example, also offers a live speech translation feature. [17]

2.4.2 AI-Agents in General

zunächst soll geklärt werden was unter KI-Agenten verstanden wird. Letztlich sind KI-Agenten systeme um teilaufgaben zu automatisieren. Um ein präziseres Verständnis zu bekommen lohnt es sich daher eine abgrenzung zu klassen workflow automatisierungen zu machen.

Programmatische Prozessautomatisierung bezeichnet die regelbasierte Automatisierung von Abläufen, bei der festgelegte Bedingungen und strukturierte Entscheidungslogiken genutzt werden, um Prozesse effizient, zuverlässig und wiederholbar auszuführen. Techniken die hier zum einsatz kommen sind robotic Process Automation, Scripting und Markos.

LLMs können kontext dabei helfen unstrukturierte aufgaben zu lösen. Anthropic hält folgende Definition bzgl. von Agenten welche KI-Agenten von Workflows abgrenzt. Workflows are systems where LLMs and tools are orchestrated through predefined code paths. Agents, on the other hand, are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.

2.4.3 Building blocks for AI-Agents

Augmented LLM

Prompt chaining

Routing

Parallelization

Orchestrator-workers

Evaluator-optimizer

2.4.4 Frameworks to design LLM-based Meeting AI-Agents

The Observe, Ask, Intervene Framework for designing AI Agents for more inclusive Meetings

In study by Houtti et al.[18, p. 5], in which participants were asked to design virtual co-hosts during semi-structured design sessions, a widespread tendency emerged to design the co-host to be interactive rather than to let it act entirely automatically. Common design patterns involved the co-host providing prompts in response to explicit instructions (e.g., when explicitly asked or when a pin was set). The authors conclude that interactivity could help prevent mistrust.

What? When? Who? Framework for Goal relecting (Are we on track?)

AI-Agents in General

2.4.5 Proactive AI-Interactions: Timing and triggers for Human-AI-Interaction-Systems

To reduce the need for explicit prompts, an AI agent should be able to proactively recognize when it is its turn in an interaction sequence. In dyadic human-AI interactions, pauses or so-called “stop words” are often used for this purpose to identify a change in speaker. In multi-party scenarios (e.g., group conversations), however, this becomes significantly more complex and leads to the “multi-party turn-taking problem”. [19]

Some approaches model the problem as a machine learning task by attempting to predict the next speaker based on previous speaker changes. [20] Bayer et al. employ traditional algorithms such as Maximum Likelihood Estimation (MLE) and Support Vector Machines (SVMs), as well as deep learning architectures including Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. They distinguish two primary paradigms: agent-only models, which rely solely on speaker history, and agent-and-content models, which additionally incorporate conversational context. Both paradigms have been trained and evaluated across diverse, topic-oriented and chit-chat corpora using these varied algorithms. The empirical findings indicate that while simple agent-only models are sufficient for small, highly focused dialogues, incorporating conversational content becomes crucial in goal- and topic-oriented interactions characterized by complex, irregular turn-taking patterns. Conversely, in unstructured chit-chat scenarios, the integration of semantic content yields no statistically significant performance gain over simple, rule-based baselines such as the “Repeat Last heuristic” [20, p. 2]

With these approaches, however, it remains unclear why an agent should interact at a specific point in time. Due to the design of the ML algorithms, the timing of the interaction is decoupled from its cause or is only implicitly included in the model, but cannot be explicitly explained. (Example: We know that it is now the time for the agent to interact now -> But why? Whats the reason its like that he should talk? What should he say?)

Die über die letzten Jahre stattgefundenen Entwicklungen bei LLMs ermöglichen neue Ansätze um diese Probleme zu modellieren. Neues Paradigma von LLM-basierten Agentischen Systemen. Was ist das? -> Hier beschreiben was KI-Agenten sind (Anthropic Definition finden und verwenden) LLM-basierte AI-Agenten mit einer LLM-Pipeline, welche intrinsische Gedanken simuliert, könnte ein viel versprechender Ansatz sein.

Liu et al. verfolgen diesen Ansatz um einen proaktiven conversational Agent mit inneren Thoughts zu modellieren. [21]

Y verfolgen diesen Ansatz um einen proaktiven Chatbot zu modellieren und schlussfolgern, dass

Classic Workflow

2.4.6 Theoretical frameworks: Human-AI Interaction (HAI), Explainable AI (XAI)

2.5 Technology Acceptance and Usability

The questionnaire instrument empirically validated by Robert Davison [22] for the systematic assessment of meeting success was used as the basis. This instrument measures the success of meetings based on five core dimensions: communication, quality of discussion, status effects (power structures and hierarchies), teamwork, and efficiency. Although the questionnaire is widely used, it should be noted that the general questions regarding technology use are based on assumptions about technologies of that time, which primarily concerned passive, computer-supported collaboration tools. They do not, however, address autonomous, proactive AI agents that simulate behavior close to that of humans. For this reason, questions regarding proactive AI systems were added to the questionnaire.



2.5.1 Technology Acceptance Model (TAM)

2.5.2 System Usability Scale (SUS) and psychological effects

3 Empirical Analysis of Facilitating Strategies (RQ1)

3.1 Methodology: Expert Interviews

To complement the findings from the literature review on the challenges in meetings—particularly in decision-making processes—with practical experience, a qualitative study was conducted. Semi-structured expert interviews were chosen as the method. This type of interview allows for in-depth insights into the respondents' perspectives through open-ended questions and flexible follow-ups, while at the same time ensuring a thematic focus through a guide.

Expert interviews are particularly suitable because experts, through their specialized knowledge and practical experience, provide efficient access to specific areas of knowledge [23, p. 877]. The definition of an expert is context-dependent and must be determined on a case-by-case basis depending on the research topic [23, p. 877]. In this study, only individuals with verifiable professional experience in the field of facilitation and who are professionally active in this context were interviewed. Therefore, the interviews conducted can be classified as semi-structured expert interviews.

Laut Cornelia Helfferich [23, p. 877] könne davon ausgegangen werden, dass Personen die für eine Interview bereit sind die Wahrheit sagen. Dies subjektive Wahrheit sei allerdings kann durch die Erzählpraktiken (z.B. ausschmücken und verkürzen) des zu interviewten gepärgt werden.

Von der Methodik unterscheidend sind Experteninterviews nicht zu

3.1.1 Sampling: Purposeful selection of 5 facilitating experts

Für diese Studie wurden 5 Experten befragt.

3.1.2 Data Collection: Conduction of semi-structured interviews (30 min)

Wann wurden die interviews geführt? Inwlechn Zeitraum? Es wurden nie mehr als 2 Interviews pro Tag geführt. Es gab einen Interviewer, nämlich den Autor dieser Arbeit. Abwohl die Interviews leicht variiert haben, haben diese im Schnitt ca. 30min gedauert. Die Interviews haben über Zoom stattgefunden und wurden, um diese später besser zu transcribieren aufgezeichnet. Jedes Interview startet mit einer kurzen Einleitung in der erklärt wurde, was der Zweck des Interviews ist, wofür die Datenerhebungen werden und wie diese verarbeitet werden. Anschließend wurde die Interviewten gefragt, ob diese mit dieser Verarbeitung und insbesondere der Aufnahme einverstanden sind. Dem haben alle zugestimmt.

Zu Beginn bereits Kontext zu liefern, hat dabei geholfen klar einzuordnen, wofür es in dem Interview gehen soll.

Through these measures and also through the choice of the interview channel, a certain closeness could be created, which is helpful in obtaining the views and opinions of the interviewee 31 (Meyen et al., 2019).

Nach dem das Interview beendet wurde, wurden sich direkt Notizen gemacht und erste Gedanken festgehalten, während die Eindrücke nach den Interviews noch frisch waren.

3.1.3 Analysis: Thematic coding

Deductive Coding kann missen auf wertvolle Einsichten. Während induktives Codieren besser für untersuchte neue Ideen oder Konzepte ist. Da allerdings bereits eine Literaturrecherche vorausgesetzt wurde, besteht zumindest ein Verständnis darüber, welche Themenkomplexe in den Interviews zu erwarten sind bzw. welche explizit abgedeckt werden sollen, um genauere Informationen von den Experten zu erlangen. Aus diesem Grund wurde ein Kategorisierungssystem basierend auf den Literaturrecherchen erstellt. Und daraufhin wurde im Anschluss allerdings ein induktives Codieren der Interviewergebnisse durchgeführt. Dieser Ansatz sollte es ermöglichen einen klaren Rahmen für die Interviews zu schaffen.

aber dennoch offengenug zu sein um neue, noch nicht in der Recherche identifizierten Informationen aufnehmen zu können.

Somit wurde ein hybrider coding ansatz gewählt. bei dem die codes iterativs verreinert wurden. Auch die überkategorien wurden dann im nachgang weiterfeinert.

3.2 Results: Facilitating Strategies and Challenges

3.2.1 Comparison with literature (Chapter 2)

3.3 Implications for AI System Design

3.3.1 Derived requirements for RQ2

4 System Design and Implementation (RQ2)

4.1 Requirements Analysis

4.1.1 Functional Requirements

-> es gibt zwei haupt unterschieden die man treffen kann: procativ system und reactive systeme. In are we on track wurden zwei verscheidene system design (aktiv und passiv) beiee hatte vor und nachteile. (Welche) Sind aber beide relevant. Daher Plattform, welche beides kann, agenda live verfolgen und agent kann selbst einschreiten.

Im allgmeien gibt es zwei ansätze ein solches system umzusetzen: 1) Intergration im bestehende Meeting-Tools wie z.B ms Zoom, Teams etc 2) Protoyp neu entwickeln mit einger video conference funtkionalität

variante ein hat den vorteil, dass sich ledeglich auf das design und die implementeirung des KI-Agent fokussiert werden muss.allerdings ist mn+an limiert durch die integrationsmöglichkeiten, welch die planfromen über die API bereitstellen. Dies varriert je nach platrfom. es ist erschwer mölglich anpasseungen an dem vorgeben UI der Konferenzen forzuhenhemn oder sogar unmöglich. somit ist der lösungsruam f+r den prototyp limiert.

Variante 2) Vorteil komplett frei, nachtiel user nicht vertraun mti interfa

4.1.2 Non-Functional Requirements

4.2 System Architecture

4.2.1 Components: Transcription, Analysis, and Intervention Modules

4.2.2 Technology Stack: NLP Libraries, Backend/Frontend Frameworks

4.3 Prototype Development

4.3.1 Implementation Details

4.3.2 Limitations of the Prototype

5 Empirical Validation (RQ2 and RQ3)

5.1 Methodology: Usability Study

5.1.1 Study Design

5.1.2 Sample

5.1.3 Data Collection

5.2 Results

5.2.1 Usability (SUS Scores and Analysis)

5.2.2 Acceptance (TAM: Perceived Usefulness, Ease of Use, Psychological Effects)

5.3 Discussion

5.3.1 Implications for System Design

5.3.2 Theoretical Contributions

5.3.3 Practical Impact for Organizations



6 Conclusion

6.1 Summary of Findings

6.2 Limitations

6.3 Future Work

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7 Appendix

7.1 Semi-Structured Interview Guide for Expert Interviews with Facilitators

This annex documents the original-language instrument (German) used in the expert interviews (Chapter 3). The guide operationalizes RQ1 (facilitating strategies and challenges in goal-oriented remote meetings with convergent objectives) and prepares the derivation of design requirements for RQ2 (a proactive, context-sensitive AI moderation system). It does not address RQ3 (user acceptance among knowledge workers), which is examined in a separate study.

The instrument follows the logic of a semi-structured expert interview: a small set of open narrative prompts (*Hauptfragen*) secures thematic focus, while optional probes (*Nachfragen*) are used only if a theoretically relevant aspect is not addressed spontaneously [23, p. 877]. The sequence may follow the conversation; not every probe needs to be asked. The intended duration is approximately 30 minutes.

Expected thematic clusters and coverage

The blocks are aligned with the challenges identified in the literature (Chapter 2) and with the system-design dimensions of RQ2 (detection, timing, addressee, and form of intervention). The following clusters are expected in the material; they serve as a sensitizing frame for later thematic coding, not as a closed coding scheme.

- C1 Goal clarity and structure.** Unclear or poorly communicated objectives; agenda without operational sub-goals; loss of a shared sense of “on track”. Covered by Block 2 and Block 3.

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- C2 Thematic deviation and time-to-decision.** Off-topic sequences, circular discussion, delayed or avoided decisions in convergent processes (decision-making, prioritization, problem-solving). Covered by Block 2 and Block 4.
- C3 Participation, hierarchy, and group dynamics.** Unequal speaking time, dominance, reluctance to dissent, face-saving. Covered by Block 2, Block 4 (public vs. private), and Block 6.
- C4 Dynamic goals and situational uncertainty.** Goals that shift during the meeting; false assumptions; need to renegotiate the mandate. Covered by Block 2 and Block 3.
- C5 Preparation versus in-meeting facilitation.** What can be secured beforehand, and what can only be steered in situ. Covered by Block 3.
- C6 Recognizing critical moments (cues for detection).** Verbal, temporal, social, and remote-specific signals from which facilitators infer that goal orientation or decision quality is at risk. Directly informs the detection logic of RQ2 (*What? When? Who?*). Covered by Block 4.
- C7 Intervention repertoire.** Methods, structures, and habits (e.g. timeboxing, roles, visualization); choice of addressee and channel (group vs. private); tone and timing. Covered by Block 3 and Block 4.
- C8 Tool support and gaps.** How current platforms help or hinder goal orientation; missing support. Covered by Block 5.
- C9 Role, agency, and limits of an AI co-facilitator.** Proactive versus on-demand support; essential functions; undesired automation; risks of unsolicited intervention (trust, control, false positives, privacy). Covered by Block 6.

Removed from the earlier draft were biographical and attitudinal items (general opinion of online collaboration, career entry, liking the role, overall rating of online vs. in-person work). They do not contribute to RQ1 or RQ2 and consume time needed for narrative accounts of practice.

Hinweise zur Durchführung

- Anrede durchgängig *Sie*; Fachbegriffe der interviewten Person aufnehmen (Moderation, Facilitation, Prozessbegleitung).

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- Hauptfragen als Erzählaufforderungen stellen, nicht als Ja/Nein- oder Bewertungsfragen.
 - Immanente Nachfragen (am Gesagten anknüpfen) haben Vorrang vor exmanenten Sondierungen.
 - Die optionalen Nachfragen sind Sensibilisierungsanker aus der Literatur; sie sollen nicht als Checkliste abgefragt werden.
 - Mit *Entscheidung* ist in diesem Leitfaden ein verbindliches Gruppenergebnis gemeint, nicht beliebige Meinungsäußerung: die Gruppe legt fest, wie es weitergeht, was priorisiert wird oder worauf sie sich als Team committet (Ziele, nächste Schritte, gemeinsame Arbeitsweisen). Diese Arbeitsdefinition bei Bedarf im Gespräch wiederholen.
 - Nach dem Gespräch: unmittelbares Interviewprotokoll (erste Eindrücke, besondere Sequenzen, Kontext) anfertigen, solange der Eindruck noch frisch ist.

Einstiegstext (sinngemäß)

Vielen Dank, dass Sie sich Zeit nehmen. Ich untersuche, wie zielorientierte Remote-Meetings so moderiert werden, dass die Gruppe beim vereinbarten Ziel bleibt und zu einem verbindlichen Ergebnis kommt, also zu einer Entscheidung im engeren Sinn.

Gemeint sind damit Entscheidungssituationen, in denen die Gruppe zu einer Festlegung gelangen muss darüber, *wie es weitergeht*. Die Art der Entscheidung kann dabei sehr vielfältig sein. Typische Fälle sind zum Beispiel:

- Priorisierung, etwa welches Vorhaben oder Feature als Nächstes umgesetzt wird.
- Commitment des Teams, etwa welche Ziele es sich setzt.
- verbindliche nächste Schritte oder die Festlegung gemeinsamer Arbeitsweisen.

Entscheidend ist, dass aus dem Meeting eine gemeinsame Festlegung folgt, an der sich die weitere Arbeit ausrichten kann.

Aus den Gesprächen mit erfahrenen Moderatorinnen und Moderatoren leite ich ab, welche Anforderungen ein KI-gestütztes Unterstützungssystem erfüllen müsste, um Teilnehmende dabei zu unterstützen, genau bei solchen Aufträgen am Ziel zu bleiben und zügiger zu einer solchen Festlegung zu kommen.

Das Gespräch ist als offenes Experteninterview angelegt. Es gibt keine richtigen oder falschen Antworten; mich interessiert Ihre berufliche Praxis und Ihre Einschätzung. Das Interview dauert etwa 30 Minuten, wird über Zoom geführt und — mit Ihrem Einverständnis — aufgezeichnet, transkribiert und in anonymisierter Form für die Bachelorarbeit ausgewertet. Die Aufnahme wird nicht an Dritte weitergegeben. Sind Sie mit Aufzeichnung und dieser Verarbeitung einverstanden?

Einverständnis dokumentieren, dann aufzeichnen.

Block 1: Beruflicher Kontext

Funktion: Sampling-Validierung und gemeinsame Bezugsebene. Ca. 3 Minuten.

- 1.1 Bitte beschreiben Sie kurz Ihre Rolle und Ihren Erfahrungshintergrund in der Moderation — insbesondere in Remote-Meetings, in denen die Gruppe zu einer verbindlichen Festlegung kommen soll (nächster Schritt, Priorisierung, Ziele oder gemeinsame Arbeitsweisen).
- *Optional:* Seit wann sind Sie in diesem Feld tätig, und in welchen organisatorischen Kontexten (z. B. Unternehmen, öffentliche Verwaltung, agiles Coaching, Großgruppen)?
 - *Optional:* Wer sind typischerweise die Teilnehmenden (Hierarchie, Fachlichkeit, Gruppengröße, wiederkehrendes Team vs. ad hoc)?

Block 2: Herausforderungen in zielorientierten Remote-Meetings (RQ1)

Funktion: Empirische Verdichtung der in Kapitel 2 identifizierten Herausforderungen; Fokus auf konvergente Prozesse, nicht auf Online-Zusammenarbeit im Allgemeinen. Ca. 6 Minuten.

- 2.1 Wenn Sie an typische Remote-Meetings denken, in denen eine Entscheidung getroffen werden soll: Was läuft dort aus Ihrer Erfahrung häufig schief — und woran merken Sie das im Verlauf des Gesprächs?
- *Falls nicht von selbst thematisiert:* Zielklarheit und Struktur; Abschweifen vom Thema; ungleiche Beteiligung oder hierarchische Dynamiken; Zeitdruck; Ziele, die sich während des Meetings verschieben.

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- 2.2 Welche dieser Schwierigkeiten sind für Sie spezifisch für das Remote-Setting — und welche würden in vergleichbarer Form auch in Präsenz auftreten?

Block 3: Strategien der Ziel- und Entscheidungssicherung (RQ1)

Funktion: Rekonstruktion bewährter Praktiken (Methoden, Strukturen, Gewohnheiten) und der Sichtbarmachung des Ziels für die Gruppe. Ca. 6 Minuten.

- 3.1 Wie gehen Sie vor, damit ein solches Meeting am Ziel bleibt und die Gruppe zu einer Entscheidung kommt? Beschreiben Sie gerne ein konkretes Vorgehen anhand einer typischen Situation.
- *Optional:* Welche Methoden, Strukturen oder Gewohnheiten haben sich für Sie bewährt (z. B. Timeboxing, Rollen, Visualisierung, Entscheidungsregeln)?
 - *Optional:* Was davon legen Sie in der Vorbereitung fest, und was steuern Sie erst in der Situation?
- 3.2 Wie stellen Sie sicher, dass den Teilnehmenden während des Meetings klar bleibt, was in diesem Moment “on track” bedeutet — also was das aktuelle Ziel bzw. der aktuelle Entscheidungsauftrag ist?
- *Optional:* Was geschieht, wenn das Ziel selbst unklar ist oder sich im Meeting als unpassend erweist?

Block 4: Kritische Momente, Erkennung und Intervention (RQ1, Anschluss an RQ2)

Funktion: Explikation impliziten Expertenwissens über Zeitpunkt, Anlass und Form von Interventionen. Die Cues sind zentral für die spätere Spezifikation der Erkennungslogik. Ca. 7 Minuten.

- 4.1 Was sind in zielorientierten Meetings mit Entscheidungsauftrag typische kritische Momente? Was kann dort konkret schiefgehen — für das Ziel, für die Entscheidung und für die Gruppe?

4.2 Woran erkennen Sie diese Momente? Können Sie das möglichst an konkreten Beobachtungen festmachen — etwa daran, wie gesprochen wird, wer spricht oder schweigt, wie die Zeit vergeht, oder woran Sie im Remote-Setting merken, dass die Gruppe den Faden verliert?

- *Optional, nicht als Messzwang formulieren:* Gibt es Anzeichen, die sich für Sie vergleichsweise zuverlässig wiederholen, sodass Sie fast schon sagen könnten: Jetzt muss ich eingreifen?

4.3 Wann sprechen Sie jemanden direkt in der Gruppe an, wann eher auf einem privaten Kanal — und nach welchen Kriterien treffen Sie diese Wahl?

- *Optional:* Was riskieren Sie, wenn Sie zu früh, zu spät, zu öffentlich oder zu privat eingreifen?

Block 5: Werkzeuge und Unterstützungsbedarf (Brücke RQ1 → RQ2)

Funktion: Bestehende sozio-technische Praxis und Lücken, ohne bereits eine KI-Lösung nahelegen. Ca. 4 Minuten.

5.1 Welche Plattformen und Werkzeuge nutzen Sie in der Moderation — und inwiefern helfen oder behindern diese Sie dabei, Zielorientierung und Entscheidungsfortschritt zu sichern?

- *Optional:* Welche Unterstützung hätten Sie sich in konkreten Situationen gewünscht, die heutige Werkzeuge nicht leisten?

Block 6: Anforderungen an ein KI-gestütztes Moderationssystem (RQ2)

Funktion: Offene Anforderungserhebung aus Expertensicht. Das System wird als Unterstützung der Teilnehmenden bei Zielbindung und Entscheidungsgeschwindigkeit eingeführt, nicht als Ersatz der menschlichen Moderation. Ca. 7 Minuten.

6.1 Stellen Sie sich ein KI-System vor, das in Remote-Meetings die Teilnehmenden dabei unterstützt, am vereinbarten Ziel zu bleiben und zügiger zu einer Entscheidung zu kommen. Wie sollte ein solches System aus Ihrer Sicht agieren — und was sollte es ausdrücklich nicht tun?

- *Optional:* Eher proaktiv von sich aus, oder nur auf Anfrage?

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- *Optional*: Welche Rolle wäre angemessen (z. B. stiller Spiegel, Zeit- und Zielwächter, Co-Moderatorin, Vorschlagsgeberin)?
- 6.2 Welche Funktionen wären für Sie essenziell, damit das System tatsächlich nützt — bezogen auf das Erkennen von Abweichungen, auf den Zeitpunkt und die Form der Intervention sowie darauf, wen das System anspricht (Gruppe oder einzelne Person)?
- *Optional*: Wenn das System zwischen öffentlicher Intervention und privater Rückmeldung wählen könnte: Nach welchen Kriterien sollte es das tun?
- 6.3 Welche Risiken, Schwächen oder unbeabsichtigten Wirkungen sehen Sie, wenn ein KI-System von sich aus in Meetings einschreitet?
- *Falls nicht von selbst thematisiert*: Vertrauen und Kontrolle der Gruppe; Fehlalarme; Gesichtswahrung; hierarchische Verzerrung; Datenschutz und Transparenz.

Abschluss

- 7.1 Gibt es Aspekte der Zielorientierung, der Entscheidungsfindung oder der KI-Unterstützung, die wir nicht angesprochen haben und die Ihnen wichtig erscheinen?

Dank, Hinweis auf Anonymisierung und — falls zugesagt — auf die Möglichkeit, Ergebnisse zurückzuspiegeln. Aufnahme beenden.